

EWAT: Early Warning and Anomaly Typing in Kubernetes Microservice Architectures

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Abstract

Kubernetes microservice deployments produce a continuous stream of *benign* change — rolling deployments, autoscaling, reconfiguration — that classical anomaly detectors confuse with genuine faults, yielding massive false-positive rates in production. We present EWAT (*Early Warning and Anomaly Typing*), a pipeline that explicitly separates benign drift from real anomalies before learning an empirical ontology of fault types. EWAT is positioned as *early warning* (*what* fault, *how long before*) and is deliberately distinct from root-cause analysis (*where/why*, post-mortem). The four-stage pipeline combines (i) drift detection by Maximum Mean Discrepancy with Random Fourier Features and a *look-through* mechanism, (ii) a spatio-temporal graph convolutional encoder, (iii) contrastive siamese typing with agglomerative clustering, and (iv) per-type precursor classifiers, capped by a formal OWL/RDF ontology built with Transfer Entropy (KSG estimator). We evaluate on data collected from a real nine-node RKE2 cluster (`observit-cluster1`) under controlled Chaos Mesh fault injection. We report results with deliberate statistical honesty. Our **defensible headline is a macro-AUROC of 0.920 (95% bootstrap CI [0.878, 0.956])** for active-scenario typing against the *independent* Chaos Mesh ground truth on the `ewat_v4_strat` dataset — a deterministic logistic-regression baseline that does *not* require the learned encoder. The latent space is geometrically structurable (H1: silhouette = 0.782 ± 0.065 over ten seeds on `ewat_v3`). We show that the much higher self-referential precursor AUROC (0.987 ± 0.011) is *circular* and, via a battery of stress tests, that genuine temporal precursion exists but only surfaces against the independent target ($\Delta_{\text{far}-\text{near}} = -0.116$). Negative results are reported as contributions: the look-through hypothesis is robustly falsified ($p = 0.27$ on `ewat_v3`, $p = 0.37$ on `ewat_v4_strat`), open-set generalisation to unseen fault types is only partial (OpenMax unknown-AUROC = 0.55), and the number of empirical types is intrinsically unstable. End-to-end inference latency is 13 ms at the 95th percentile, $375\times$ under a 5 s budget.

Keywords: anomaly detection, early warning, microservices, Kubernetes, STGCN, Transfer Entropy, concept drift, contrastive typing, open-set recognition.

1 Introduction

Microservice architectures deployed on Kubernetes are in permanent flux. Rolling deployments, horizontal autoscaling, configuration changes and traffic ramps all shift the statistical distribution of observability signals without any underlying fault. Anomaly detectors that react to distributional change therefore fire on these *benign drifts* exactly as they fire on real incidents. We confirm this directly: a per-feature z -score detector raises a false alarm on 100% of benign-drift episodes at every threshold $\sigma \in [2.0, 3.5]$ we tested (Section 6). The conflation of drift and anomaly is the operational problem EWAT attacks.

Early warning, not RCA. EWAT is an *early-warning* system: it predicts *what* type of fault is developing and *how long before* it manifests. It is not root-cause analysis (RCA). RCA is post-mortem and answers *where* and *why* a failure occurred; EWAT acts *before* the failure and answers *what* and *when*. Table 1 makes the distinction explicit. This boundary is a design commitment, not a slogan: the typing and ontology stages approach the “what” but never attribute a root cause, and our evaluation metrics (lead time, false-alarm rate on drifts) are warning metrics, not diagnostic ones.

Four operational regimes. A central modelling choice is that we recognise *four* regimes, not three:

Table 1: EWAT (early warning) versus root-cause analysis.

	EWAT (early warn- ing)	RCA
Question	<i>What / when</i>	<i>Where / why</i>
Timing	before the fault	after the fault
Output	typed precursor alert	causal explanation
Metric	lead time, FA on drift	localisation accuracy

$\theta \in \{\theta_{\text{normal}}, \theta_{\text{drift}}, \theta_{\text{anomaly}}, \theta_{\text{drift} \cap \text{anomaly}}\}$. The fourth regime $\theta_{\text{drift} \cap \text{anomaly}}$ models the faulty-deployment case in which a benign drift and a real anomaly occur simultaneously — precisely the case where naive detectors are most dangerous. The look-through mechanism (Stage 0) is designed to keep the signal flowing during drift rather than zeroing it out, so that an anomaly hidden inside a drift remains observable.

Contributions.

1. An atomic, reproducible four-stage early-warning pipeline (Section 4), implemented as independent testable modules (425+ unit tests) on a real cluster.
2. A *geometric* contribution validated as H1: the STGCN latent space is structurable into separable fault types, silhouette = 0.782 ± 0.065 over ten seeds (Section 6), driven by the cosine-average clustering choice on the L2-normalised embedding

sphere.

3. A *statistically honest* predictive headline: macro-AUROC = 0.920 [0.878, 0.956] for active-scenario typing against the independent Chaos Mesh ground truth, deterministic and without the learned encoder.
4. Evidence, via a stress-test battery (Section 7), that the self-referential precursor AUROC (0.987) is *circular*, while genuine temporal precursion exists against the independent target ($\Delta_{\text{far-near}} = -0.116$).
5. Reported *negative* results as contributions: look-through falsified twice ($p = 0.27$, $p = 0.37$), partial open-set recognition (OpenMax 0.55), unstable type count.
6. A formal OWL/RDF ontology of fault types (29 literature-anchored classes, 143 individuals, HermiT-consistent) extracted with multivariate Transfer Entropy (KSG), never Granger causality.

All numbers in this paper are traceable to the project’s experiment logs; we name the dataset and configuration for every figure, and we surface — rather than hide — the residual inconsistencies we found between intermediate runs (Section 8).

2 Related Work

RCA for microservices. The dominant body of work on microservice reliability targets root-cause analysis. Fu et al. [1] survey the field and identify a persistent gap between benchmark performance and production deployability, driven largely by false positives. Soldani and Brogi [2] catalogue microservice anti-patterns and failure modes; we reuse their taxonomy, together with Gregg’s USE methodology [3] and cascading-failure modelling [4], to anchor our OWL class hierarchy in the literature rather than in ad-hoc cluster labels. EWAT departs from this line of work in objective: we predict and type developing faults rather than explaining past ones.

Concept drift. Distinguishing benign distributional change from genuine anomalies is a concept-drift problem. Hinder et al. [5] formalise drift in unsupervised streams, and Myrtollari et al. [6] propose drift-aware anomaly detection specifically for Kubernetes microservices. EWAT operationalises the drift/anomaly separation as an explicit fourth regime $\theta_{\text{drift} \cap \text{anomaly}}$ and a look-through filter built on the kernel two-sample test [7], estimated cheaply with Random Fourier Features. Our negative result on look-through (Section 7) is, to our knowledge, a useful data point on the limits of MMD-based separation on short episodes.

Representation learning on service graphs. We encode the time-evolving service graph with a spatio-temporal graph convolutional network (STGCN) [8],

and compare against graph attention [9] and contrastive pre-training in the SimCLR style [10]. Type structure is assessed with the silhouette criterion, using the conventional acceptability threshold of 0.3 [11], complemented by the gap statistic [12] for choosing the number of clusters.

Causality and interpretability. For the ontology we estimate directed information flow with Transfer Entropy using the KSG estimator [13] — never Granger causality, which assumes linearity inappropriate for our signals — and control multiple testing with Benjamini–Hochberg FDR [14]. Feature attributions are validated with SHAP [15] after we found gradient-based saliency to be unreliable ($\rho = -0.34$ against permutation importance).

Open-set recognition. Because production faults include types never seen in training, we evaluate open-set recognition with OpenMax [16] and discuss Mahalanobis [17] and energy-based [18] alternatives as future work. For external validation we adapt the public RCAEval RE2-OB benchmark [19].

3 Problem Formalisation

3.1 Service graph

We model the system as a time-evolving weighted graph $\mathbf{G}(t) = (V, E(t), w_E(t))$. Vertices V are Kubernetes *services* and *deployments* (not pods), with $|V| = N$ held constant over an episode. Each edge carries a vector weight $w_E(t) : E(t) \rightarrow \mathbb{R}^3$, $e_{ij}(t) \mapsto (\text{volume}_{ij}, \text{latency}_{ij}^{\text{med}}, \text{error-rate}_{ij})$; an edge is present when its call volume exceeds zero over the sliding window.

3.2 Telemetry signal

The per-step signal stacks three modalities,

$$\mathbf{S}(t) = [\mathbf{M}(t) \mid \mathbf{T}(t) \mid \mathbf{L}(t)] \in \mathbb{R}^{N \times 17}, \quad (1)$$

with $\mathbf{M}(t) \in \mathbb{R}^{N \times 7}$ metrics (CPU, RAM, latency P99, HTTP error rate, network saturation, disk I/O, queue depth), $\mathbf{T}(t) \in \mathbb{R}^{N \times 6}$ trace features (span-duration P99, abnormal-span rate, trace depth, fan-out, retry rate, latency coefficient of variation), and $\mathbf{L}(t) \in \mathbb{R}^{N \times 4}$ log features (error rate, warning rate, semantic anomaly score as mean SentenceBERT distance to the normal centroid, lexical entropy). Metrics come from the existing Prometheus stack; traces and logs from the OpenTelemetry collector. Crucially, $\mathbf{S}(t) \sim D_{\theta(t)}(\mathbf{G}(t))$ is *not* an additive mixture of regime effects.

Differentiated aggregation. Intra-service aggregation is component-specific, never a plain mean: saturation components (CPU, RAM, net, disk I/O) aggregate by *max*; rate components (error, warning) by *volume-weighted sum*; latency components (P99, span duration)

Table 2: Regime θ to dataset label correspondence (labels.parquet).

Label	Regime
normal	θ_{normal} (pre-injection reference)
injection	θ_{anomaly} (chaos active)
drift_anomaly	$\theta_{\text{drift} \cap \text{anomaly}}$ (simultaneous faulty deploy)
recovery	post-injection return (excluded from H1/H3)
drift_flag	θ_{drift} (detected, not a label)

by the *P99 of the union* of raw per-pod distributions (never a percentile of percentiles); structural components (trace depth, fan-out, lexical entropy) by *median*.

3.3 Operational regimes

The regime variable is $\theta(t) \in \{\theta_{\text{normal}}, \theta_{\text{drift}}, \theta_{\text{anomaly}}, \theta_{\text{drift} \cap \text{anomaly}}\}$. Table 2 gives the correspondence to dataset labels. Benign drift θ_{drift} (rolling deploy, autoscaling) is *not* a regime label; it is detected orthogonally by the drift detector through the boolean **drift_flag**. The combined regime $\theta_{\text{drift} \cap \text{anomaly}}$ is materialised by the **faulty_deploy_overlap** scenario.

3.4 Hypotheses and falsification

EWAT is organised around four falsifiable hypotheses. **H1 (structurability)**. Fault types form separable groups in the latent space; falsified if held-out silhouette < 0.3 [11] across seeds and splits. **H2a (drift separability)**. Look-through reduces the false-positive rate at constant recall; falsified if no significant reduction ($p > 0.05$). **H2b (combined-regime identification)**. The pipeline identifies $\theta_{\text{drift} \cap \text{anomaly}}$ as distinct from pure drift and pure anomaly; falsified if no cluster shows simultaneous **drift_flag** and precursor alert above chance. **H3 (predictability)**. Per-type precursors beat the 0.5 baseline for some horizon k ; falsified if $\text{AUROC} < 0.5 \forall k$. We additionally distinguish *circular* H3 (self-referential cluster labels as target) from *honest* H3 (independent Chaos Mesh scenarios as target), a distinction that turns out to be decisive (Sections 6 and 7).

4 Method

EWAT is a cascade of four stages (Figure 1); stages 0, 1 and 3 run online within a latency budget, while typing and ontology construction (stage 2/2b) run offline.

4.1 Stage 0 — Drift detection (MMD-RFF, look-through)

We estimate $\text{MMD}^2(W_{\text{ref}}, W_{\text{cur}})$ between a reference and a current window with Random Fourier Features $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}^D$, giving $O(nD)$ cost. The *look-through* filter never zeroes the signal during drift: if $\text{MMD}^2 <$

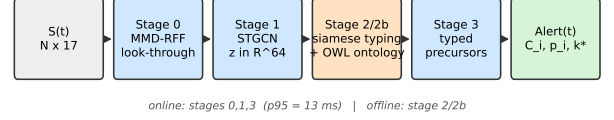


Figure 1: The EWAT four-stage pipeline. Stage 0 (drift), stage 1 (encoder) and stage 3 (precursors) are online; stage 2/2b (typing and ontology) are offline.

$\varepsilon_{\text{drift}}$ the signal passes unchanged; if it exceeds the threshold and a post-drift confirmation test is positive, the signal passes flagged **DRIFT**; if the confirmation is negative, the detector recalibrates ($W_{\text{ref}} \leftarrow W_{\text{cur}}$). The threshold is calibrated by injecting benign drifts; we obtain $\varepsilon_{\text{drift}} = 0.5226$ (Youden-optimal).

4.2 Stage 1 — STGCN encoder

The encoder maps a windowed signal and graph to an embedding $z_e = \text{Enc}_\theta(\hat{\mathbf{S}}_{[t-W, t+\delta]}, \mathbf{G}(t)) \in \mathbb{R}^{64}$, combining graph convolution over the weighted adjacency with a causal temporal-convolution stack and a masked mean-pooling head (padding excluded). It is pre-trained by self-supervised reconstruction without scenario labels. The graph convolution uses the time-varying adjacency $A(t)$ (the earlier static A_{mean} implementation is retained only as an ablation variant).

4.3 Stage 2 — Contrastive typing

A siamese projection head ($d_{\text{proj}} = 64$) trained with a contrastive hinge loss (margin 2.0, semi-hard negative mining) maps embeddings onto the L2-normalised unit sphere. Because the embeddings live on a sphere, we cluster with *average-linkage agglomerative clustering under cosine distance* — aligning the metric with the geometry, a change that proves to be the principal driver of H1 (Section 6). Cross-split labels are assigned by *nearest centroid* from the training centroids, so cluster identities are consistent across train/val/test. Per-type feature attributions use permutation importance, cross-validated with KernelSHAP [15].

4.4 Stage 2b — Ontology

The ontology $O = (C, R)$ carries three relation families: temporal transitions $C_i \rightarrow^{\Delta t, \sigma} C_j$; *causal* edges from Transfer Entropy with the KSG estimator [13] (multivariate on synthesised cascades; permutation-bootstrap significance with Benjamini–Hochberg FDR [14]); and co-occurrence edges by a 2×2 Yates χ^2 with a Fisher-exact fallback and Holm correction. The result is exported as a formal OWL/RDF artefact (TBox of 29 literature-anchored classes; ABox of 143 individuals) and checked for consistency with the HermiT reasoner.

Table 3: End-to-end inference latency (CPU, 200 iterations, `ewat_v3`).

Stage	median	p95	budget
0 — drift	0.01 ms	0.01 ms	1000 ms
1+2 — en-coder+siamese	0.96 ms	1.97 ms	2000 ms
3 — precursors	1.85 ms	3.91 ms	1000 ms
Total	9.26 ms	13.28 ms	5000 ms

4.5 Stage 3 — Typed precursors

For each type C_i a one-vs-rest classifier produces a precursor probability $\hat{p}_i(t) = f_i(\hat{\mathbf{S}}_{[t-k, t]}, \mathbf{G}(t)) \in [0, 1]$. The horizon is chosen *on the validation set*, $k_i^* = \arg \max_k \text{AUROC}(f_i, k)$ over $k \in \{1, \dots, 20\}$ steps; test AUROC is reported only at k_i^* . The final alert is $\text{Alert}(t) = (C_i, \hat{p}_i(t), k_i^*, \text{card}_{C_i})$.

4.6 Latency budget

The formal budget is stage 0 < 1 s, stage 1+2 < 2 s, stage 3 < 1 s, total < 5 s. Measured on CPU over 200 iterations (Table 3), the end-to-end 95th percentile is 13.28 ms — 375× under budget.

5 Experimental Setup

Cluster and observability. All data were collected on `observit-cluster1`, a real nine-node RKE2 cluster (Kubernetes v1.32.7) running the Online Boutique demo. Two observability stacks coexist: Prometheus/Grafana for mature metrics and an OpenTelemetry collector for traces and logs, correlated through OTel semantic conventions. The service set is fixed at $N = 6$ (`frontend`, `cart`, `recommendation`, `ad`, `product-catalog`, `load-generator`); services invisible on any modality were excluded to avoid systemic NaNs.

Fault injection. Faults are injected with Chaos Mesh over 15 scenarios: 4 benign drifts (`drift_config_change`, `drift_rolling_deploy`, `drift_scale_up`, `drift_traffic_ramp`) and 11 anomalies, including `faulty_deploy_overlap` which realises the combined regime $\theta_{\text{drift} \cap \text{anomaly}}$. Each episode follows the *record* → *build* → *assemble* protocol: raw dumps are immutable ground truth, features are recomputed offline, and datasets are split last.

Datasets. Table 4 summarises the four datasets. We highlight one methodological pitfall: the original temporal split of `ewat_v4` left four scenarios entirely absent from the training set, collapsing the independent-target AUROC to a trivial 0.5. We therefore assembled `ewat_v4_strat` (stratified, at least one episode per scenario per split), which is the dataset behind our defensible headline.

Table 4: Datasets. “strat” = stratified split guaranteeing every scenario in every split.

Dataset	Episodes / split	Notes
<code>ewat_v3</code>	299;	~21 steps; disk_io 16.7% NaN
	209/45/45	(strat)
<code>ewat_v4</code>	375;	47–51 steps; 4 scen. absent train
	262/56/57	(temp.)
<code>ewat_v4_strat</code>	270/60/45	Headline dataset (strat)
<code>ewat_rcaeval</code>	90;	adapted RCAEval RE2-OB
	30	fault
		types

Protocol and methodology corrections. Unless stated otherwise, H1/H3 on `ewat_v3` use the optimised configuration (average+cosine clustering, $d_{\text{proj}} = 64$, margin 2.0, tuned logistic regression) and are aggregated over ten seeds. Two methodological corrections apply throughout: (i) H1 silhouette on val/test is computed with *nearest-centroid* labels from the training centroids, not an independent per-split clustering (which inflated scores); (ii) the precursor horizon k^* is selected on validation, never on test. Confidence intervals are 95% bootstrap (BCa for AUROC). We name the dataset and configuration for every reported number, because they differ substantially between `ewat_v3` and `ewat_v4_strat`.

6 Results

We organise results by hypothesis, then report baselines, ablations, architecture comparison, ontology, and operational alerting. Throughout we separate the *circular* evaluation target (EWAT’s own cluster labels) from the *independent* target (Chaos Mesh scenarios); only the latter supports a defensible predictive claim.

6.1 H1 — Structurability (PASS)

On `ewat_v3` with the optimised configuration, the held-out silhouette is 0.782 ± 0.065 over ten seeds (minimum 0.618, all $\gg 0.3$): H1 passes robustly. The optimised configuration improves on the initial baseline (Ward+Euclidean, 0.519 ± 0.092) by +51%; the gain is geometric and comes almost entirely from matching the clustering metric (cosine) to the L2-normalised embedding sphere. On the longer `ewat_v4_strat` episodes the silhouette is lower and more variable (0.691 ± 0.115 , range [0.521, 0.839]), and the number of types K is intrinsically unstable: neither silhouette nor the gap statistic stabilises it (modes $K = 14$ and $K = 12$ respectively, agreement on only 4/10 seeds). Figure 2 shows the per-seed distributions.

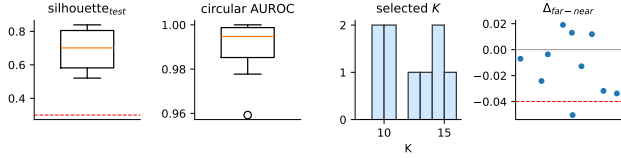


Figure 2: Multi-seed distributions (`ewat_v4_strat`, Phase H, 10 seeds): silhouette, circular precursor AUROC, selected K , and the distant-window gap Δ . The circular AUROC is stable; silhouette, K and Δ are not, and seed 42 is an outlier on Δ .

6.2 H2a — Drift separability by look-through (FAIL)

Look-through does not reduce the false-positive rate. On `ewat_v3`, the look-through detector reaches a drift true-positive rate of 0.42 versus 0.67 for a plain threshold, and an anomaly false-positive rate of 0.67 versus 0.73 — not a significant reduction ($p = 0.27$, paired one-sided t). The same test on STGCN embeddings is also negative ($p = 0.978$). Critically, the negative result is *robust* to episode length: re-running on the $2\times$ longer `ewat_v4_strat` episodes still fails (TPR = 0.50, $p = 0.372$). The mechanism is falsified, not merely under-powered; we report this as an honest negative result.

6.3 H2b — Combined regime $\theta_{\text{drift} \cap \text{anomaly}}$ (PASS but trivial)

The formal criterion passes (overlap > 30% everywhere), but trivially: the drift detector’s 5-step window fires on almost every short episode, so overlap is high regardless. A strict Fisher test of the overlap cluster (`faulty_deploy_overlap`) against pure drift gives OR = 1.48, $p = 0.35$ — not significant. Moreover, the precursor alert *precedes* the drift flag in 85–100% of cases: the drift detector is a *late* indicator, and the early warning comes from stage 3, not stage 0. H2b thus reinforces H2a rather than standing on its own.

6.4 H3 — Predictability

Circular target (self-referential). Predicting EWAT’s own cluster labels from the embedding gives a peak AUROC of 0.987 ± 0.011 (10/10 seeds PASS). We flag this as *circular*: the target is produced by the same encoder/clustering being evaluated. Section 7 shows directly that this number does not reflect temporal precursion.

Independent target (honest). Against the Chaos Mesh ground truth, our defensible headline is a deterministic logistic-regression one-vs-rest classifier on instance-normalised pre-injection features (no learned encoder): **macro-AUROC** = 0.920, 95% **bootstrap CI** [0.878, 0.956] on `ewat_v4_strat`, stable under leave-one-scenario-out (0.930 ± 0.007 , 15 folds). Table 5 consolidates all baselines by target. Two facts stand

Table 5: Baselines grouped by evaluation target. Circular = EWAT’s own labels; Independent = Chaos Mesh scenarios. CIs are 95% bootstrap.

Model (dataset)	macro-AUROC
<i>Circular target — EWAT cluster labels</i>	
Random	0.500
Raw features $\rightarrow$ EWAT labels (v3)	0.966
k -means + LR $\rightarrow$ EWAT labels (v3)	0.975
EWAT precursors, self (v3)	0.951
EWAT precursors, optimised peak (v3, 10s)	0.987 ± 0.011
<i>Independent target — Chaos Mesh scenarios</i>	
Raw features, $k=6$ (v3)	0.835 [0.773, 0.888]
STGCN z_e (v3)	0.835 [0.772, 0.885]
STGCN end-to-end (v4_strat)	0.863 [0.823, 0.905]
Raw, instance-norm best (v4_strat)	0.941 [0.909, 0.970]
Raw, inst-norm flatten LR (v4_strat)	0.920 [0.878, 0.956]

out: the encoder does *not* add aggregate predictive value (training an STGCN end-to-end on the same independent target yields 0.863, *below* the linear baseline), and the circular numbers (0.95–0.99) sit far above the honest ones (0.84–0.94).

Per-scenario redistribution. On `ewat_v3` the aggregate gap between raw features and STGCN embeddings is exactly $\Delta_{\text{macro}} = 0.000$. This is not neutrality by coincidence alone: AUROC on $n = 45$ test is a ratio of concordant pairs over a fixed integer denominator, and the encoder *redistributes* discriminability across scenarios (e.g. +27 pp on `fail_slow_cpu`, −25 pp on `noisy_neighbor`) with the per-scenario gains and losses summing to zero. The redistribution is real; its exact cancellation is an artefact of the small test set.

6.5 Ablations

Modality (rigorous, full retraining). Retraining the encoder and siamese head per condition (seed 42, `ewat_v3`), metrics-only beats the full model on H1 silhouette (0.497 vs 0.439, $\Delta = +0.058$); adding traces and logs injects geometric noise into the clustering on $n = 209$. Logs-only collapses (0.051). The hierarchy is $M > T > L$ for clustering geometry.¹

Modality (precursor H3, inference masking). For predictability the ordering inverts: the full model is best (0.954), and dropping traces and logs hurts (M -only 0.756, L -only 0.488). Traces and logs are noise for clustering geometry but *necessary* for prediction. The single most critical feature for H3 is `disk_io` ($\Delta = -0.088$ leave-one-out), despite its 16.7% NaN rate — a direct argument for the

¹An inference-time masking ablation (no retraining) gives a lower full baseline of 0.333 but the same qualitative ranking; we report the rigorous retraining figures here.

Table 6: Encoder comparison (`ewat_v3`, seed 42). AUROC is the circular precursor metric.

Encoder	K	sil_{test}	types	AUROC
STGCN	10	0.414	8/10	0.954
SimCLR	15	0.429	11/15	0.964
GAT	15	0.497	13/15	0.929

`ewat_v4` re-collection. Two feature pairs are redundant ($|\rho| > 0.9$): `latency_p99 $\leftrightarrow$ span_dur_p99` and `error_rate_http $\leftrightarrow$ abnormal_span_rate`.

6.6 Encoder architecture comparison

Table 6 compares encoders on `ewat_v3` (seed 42, corrected methodology). GAT gives the best clustering geometry and type coverage; SimCLR the best circular AUROC; STGCN the most stable K . We retain STGCN as the reference (simplest, multi-seed evidence available), while noting that none of these encoders improves the *independent*-target prediction (Section 7).

6.7 Ontology

The OWL/RDF ontology comprises 29 literature-anchored classes (TBox) and 143 individuals (ABox). Multivariate TE on synthesised cascades yields 3 significant causal edges (BH-FDR $p < 0.05$; e.g. `crash  $\rightarrow$  traffic-ramp`, TE = 0.182, $p = 0.015$), 19 co-occurrence edges, and 46 service-propagation edges after a specificity filter that drops ubiquitous traffic edges. Notably, the two benign-drift clusters produce *zero* propagation edges — a validating result, since benign drifts should not cascade. HermiT verifies consistency in 0.61 s with no inconsistent class. Eight of ten validation criteria pass; the two misses are the causal-edge count (3 vs a target of 15, limited by the synthetic corpus size) and materialised inferences (an owlready2 limitation, mitigated via SPARQL).

6.8 Operational alerting

The practical value is false-alarm control. A z -score detector raises alarms on 100% of benign-drift episodes at every σ . EWAT at threshold 0.7 cuts the drift false-alarm rate to 8.3% while keeping a 3.0 min lead time (Table 7). Lower thresholds detect more but the drift detector’s short warm-up on ~ 21 -step episodes pushes false alarms back to 100% — a data-length limitation (L1), not a design flaw.

7 Validity and Stress Tests

The circular precursor AUROC (0.987) is high enough to be suspicious. This section reports the battery of stress tests we ran to determine what that number actually measures. The headline finding is a clean dissociation: on the self-referential target the model learns the *static signature* of a scenario, not temporal precursion; on the

Table 7: Online alerting (`ewat_v3` test, 45 episodes).

Method	Detect.	Correct type	FA drift	Lead
z -score ($\sigma=2-3.5$)	100%	—	100%	2.5 min
EWAT @0.3	100%	42.4%	100%	4.6 min
EWAT @0.5	78.8%	63.6%	100%	3.9 min
EWAT @0.7	57.6%	51.5%	8.3%	3.0 min

Table 8: Stress tests on the self-referential target (`ewat_v3`).

Test	Result	Reading
A1 distant-window	$\Delta = -0.007$	no precursion (leak)
A3 permutation	$p < 0.01$	signal is real
A2 LO SO top-1	0.51 ± 0.38	no open-set generalisation
A5 paired Δ	$[-0.031, +0.044]$	encoder adds nothing

independent target, genuine temporal precursion does appear.

7.1 Self-referential target: signature, not precursion

We probe the circular evaluation with four tests (Table 8). **A1 (distant-window)**. Sliding the inference window anywhere inside the normal regime leaves the AUROC unchanged: $\Delta_{\text{far-near}} = -0.007$. If the score were precursive, moving the window earlier should degrade it. It does not — the classifier reads a static scenario signature recoverable from any point in the normal regime. **A3 (permutation null)**. The signal is nonetheless real: against a label-permutation null (0.49 ± 0.10), the observed AUROC of 0.893 gives $p < 0.01$. **A2 (leave-one-scenario-out)**. The model does not generalise to an unseen fault type: top-1 accuracy on the held-out scenario is 0.51 ± 0.38 , polarised into 100% and 0% cases — interpolation between known scenarios, not extrapolation. **A5 (paired Δ)**. The encoder adds nothing in aggregate: the paired bootstrap CI on the STGCN-minus-raw difference is $[-0.031, +0.044]$, comfortably containing zero.

7.2 Independent target: precursion reappears

The picture changes once the target is the independent Chaos Mesh ground truth. **B1 (instance-norm diagnostic)**. With global normalisation, the distant-window gap on the independent target is $\Delta_{\text{far-near}} = -0.071$ on `ewat_v3` and -0.043 on `ewat_v4_strat`: there *is* pre-injection dynamics, worth several AUROC points,

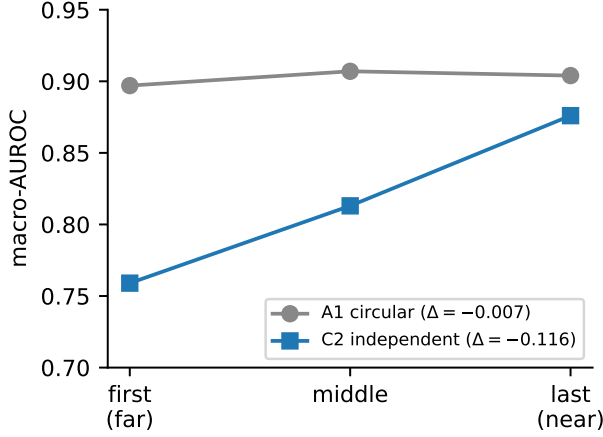


Figure 3: Distant-window analysis. On the circular target (A1) the gap is ≈ 0 (leak); on the independent end-to-end model (C2) AUROC falls as the window moves away from injection, $\Delta_{\text{far} \rightarrow \text{near}} = -0.116$ — genuine temporal precursion.

that the circular evaluation hid. Instance normalisation, which removes per-service baseline offsets, improves the signal (up to +5.5 pp at the earliest window). **B2/B3 (headline).** The resulting linear classifier reaches 0.920 [0.878, 0.956] on `ewat_v4_strat`, up from 0.855 on `ewat_v3`; the longer episodes amplify the pre-injection signal.

7.3 End-to-end model on the independent target

C1. An STGCN trained directly on the independent target reaches 0.863 [0.823, 0.905] — below the linear baseline. Consistent with A5, the encoder is not predictively useful in aggregate. **C2 (the key reversal).** Yet the *same* end-to-end model exhibits genuine temporal precursion: AUROC rises from 0.759 at the earliest window to 0.876 just before injection, $\Delta_{\text{far} \rightarrow \text{near}} = -0.116$ (Figure 3). Twelve points of AUROC depend on *when* the window sits. The leak seen in A1 was an artefact of the circular labels, not a property of the signal: against an independent target, EWAT is more than a static-signature identifier. **C3 (open-set).** OpenMax provides only partial novelty detection: unknown-AUROC = 0.55 ± 0.24 (below the 0.7 target), although top-1 unknown rate rises from 0 (closed one-vs-rest) to 0.40. We report this as a partial, honest result and discuss stronger detectors in Section 9. **C5.** The look-through retest on longer episodes confirms H2a’s failure ($p = 0.372$).

7.4 Synthesis and reproducibility caveat

The encoder’s value is *geometric* (it structures the latent space for typing, H1) and *ontological* (it grounds the OWL artefact), not aggregate-predictive. The honest predictive contribution is active-scenario typing at 0.920 with confirmed temporal precursion. A multi-seed audit

Table 9: Structural limitations and proposed fixes.

ID	Limitation	Fix
L1	short episodes (~ 21 steps)	v4/v5 long episodes
L9	H3 circularity	independent target
L10	siamese overfit / K unstable	fixed K / HDBSCAN
L13	$N = 6$, one topology	Train Ticket (v5)
L14	no cross-cluster validation	few-shot domain adapt.
L17	ontology not SRE-audited	expert review

(Phase H/J/K) tempers earlier single-seed optimism: a prior run on seed 42 reported silhouette 0.84 and $\Delta = -0.05$, but neither reproduces across ten seeds (silhouette 0.691 ± 0.115 ; Δ leak-confirmed on 9/10 seeds, seed 42 the lone outlier). The headline 0.920 is unaffected because the underlying logistic regression is deterministic; the bootstrap CI captures its only stochasticity.

8 Limitations

We documented seventeen limitations (L1–L17); Table 9 lists the structural ones with their proposed fixes. Four deserve emphasis.

Short episodes (L1) are the dominant data constraint. At ~ 21 steps, `ewat_v3` leaves too little window for the look-through warm-up, which is the proximate cause of H2a’s failure, H2b’s triviality, and the high false-alarm rate at low thresholds. `ewat_v4` (47–51 steps) mitigates but does not eliminate it (H2a still fails).

Evaluation circularity (L9) is the methodological core of this paper. The 0.987 precursor AUROC targets EWAT’s own labels; we therefore demote it and adopt the independent Chaos Mesh target (0.920) as the headline, with the stress tests of Section 7 as the evidence.

Type-count instability (L10). The siamese head overfits in 2–7 epochs on `ewat_v4` and K ranges from 9 to 15 across seeds; this is structural (small training set, diverse episode lengths), not a mining bug. We recommend fixing K or moving to density-based clustering in future iterations.

Scope and external validity (L13, L14, L17). All experiments use one six-service topology on one cluster. Zero-shot transfer to RCAEval recovers generic anomaly detection (H1 silhouette 0.684) but not type discrimination (H3 ≈ 0.5); the bottleneck is a non-transferable scaler. The ontology, while Hermit-consistent and literature-anchored, has not been audited by an operational SRE, and its causal edges are extracted from synthetic cascades.

We also note minor caveats for transparency: a single train/val/test split (no k -fold), no a priori power analysis (post-hoc: 5/10 clusters are statistically reportable at $n_{\text{pos}} \geq 5$), and stress tests A1–A5 designed *post hoc* rather than pre-registered.

9 Discussion and Future Work

What EWAT is, and is not. EWAT is a working early-warning pipeline whose honest, independently-validated contribution is active-scenario typing at 0.920 macro-AUROC with confirmed temporal precursion ($\Delta = -0.116$). It is not a generic event predictor: the learned encoder adds geometric and ontological value, not aggregate predictive power, and open-set generalisation to unseen fault types remains only partially solved. We consider the negative and nuanced results — a robustly falsified look-through, a circular metric demoted in favour of an independent one, an unstable type count — to be as much a contribution as the positive headline, because they delimit precisely where the approach is and is not trustworthy.

Roadmap. Four research axes follow directly from the limitations. *Axis A (ontology $\leftrightarrow$ prediction coupling)* would feed the causal graph back into the precursor as upstream features or a post-hoc re-ranker, testing whether the ontology adds predictive value rather than only descriptive value. *Axis B (robust temporal precursion)* would replace the STGCN with a temporal transformer, multi-seed the C2 distant-window result, and validate lead time cross-dataset on RCAEval. *Axis C (open-set)* would extend the partial OpenMax result with Mahalanobis-based and energy-based detectors and few-shot meta-learning. *Axis D (deployment)* would specify the streaming pipeline, the retraining trigger, and a formal alert contract.

Towards ewat_v5. The most consequential next step is data. ewat_v5 abandons the six-service Online Boutique for *Train Ticket* (41 Spring Cloud services), with a much larger episode budget and, crucially, a held-out set drawn from the project’s documented *real* bugs (the F1–F12 catalogue) — a literature gold standard for RCA-style evaluation. This directly attacks L1 (episode length), L13 (topology scale), L14 (cross-system validity), and L17 (real rather than synthetic faults), and would let us re-test every hypothesis on a substantially harder and more realistic system.

10 Conclusion

EWAT shows that benign drift and genuine anomalies can be separated and that developing faults can be typed ahead of time in a real Kubernetes microservice environment, provided the four operational regimes — including the combined $\theta_{\text{drift} \cap \text{anomaly}}$ — are modelled explicitly. Our two robust positive results are geometric structurability (H1, silhouette 0.782 ± 0.065 over ten seeds) and, against an *independent* ground truth, active-scenario typing at macro-AUROC 0.920 [0.878, 0.956] with confirmed temporal precursion ($\Delta_{\text{far} - \text{near}} = -0.116$) and a 13 ms p95 latency. We deliberately refuse the inflated self-referential number (0.987) and the single-seed optimism it came with, and we report the look-through

falsification and the partial open-set result as honest negatives that constrain future work rather than embarrassments to be hidden. The path forward is more and harder data: ewat_v5 on Train Ticket with real, documented faults will let us re-test every hypothesis at production scale.

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A Per-cluster precursor results

Table 10 gives the per-type precursor results on `ewat_v3` (seed 42, k^* selected on validation, AUROC on test with 95% bootstrap CI). These are *circular*-target numbers (Section 6) and should be read alongside the stress-test caveats of Section 7. Clusters with fewer than two positive test episodes (C6, C9) yield undefined AUROC and are marked n/a.

The dominant optimal horizon is $k^* = 6$ steps (3 min), the practical zone of predictability for most fault types in this dataset; C3 has an immediate signature ($k^* = 2$) and C8 (the combined-regime overlap) a slow one ($k^* = 10$).

B Cluster–scenario alignment

Normalised mutual information between clusters and Chaos Mesh scenarios is 0.518 and mean purity 0.503 —

Table 10: Per-cluster precursors (`ewat_v3`, circular target). k^* in 30 s steps.

Type	n_{pos}	k^*	AUROC	95% CI
C0	8	6	0.973	[0.906, 1.000]
C1	3	6	0.992	[0.953, 1.000]
C2	5	6	0.945	[0.865, 1.000]
C3	3	2	0.794	[0.636, 0.930]
C4	8	2	1.000	[1.000, 1.000]
C5	2	6	0.977	[0.909, 1.000]
C6	1	2	n/a	—
C7	7	6	0.992	[0.966, 1.000]
C8	7	10	0.962	[0.895, 1.000]
C9	1	2	n/a	—

the moderate alignment expected of unsupervised clustering that compresses 15 scenarios into ≈ 10 empirical types. Two scenarios sharing a latent fault signature (e.g. crash and OOM near the injection point) are a finding, not an error. Figure 4 shows the full alignment. Gradient $\times$ input saliency was found unreliable ($\rho = -0.34$ versus permutation importance) and is not used; permutation importance is cross-validated by KernelSHAP ($\rho > 0$ on 9/10 clusters).

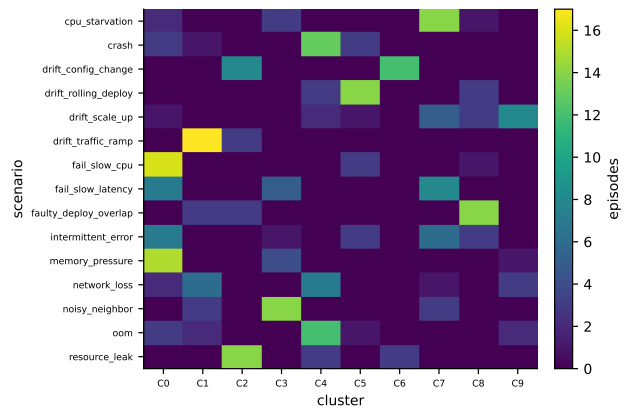


Figure 4: Scenario $\times$ cluster alignment heatmap (`ewat_v3`).